Pluralism and Alignment

SE Emerging Trends in Spatial Data Science (2026W)

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Literature

Position: A Roadmap to Pluralistic Alignment

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Abstract

With increased power and prevalence of AI systems, it is ever more critical that AI systems are designed to serve *all*, i.e., people with diverse values and perspectives. However, aligning models to serve *pluralistic* human values remains an open research question. In this piece, we propose a roadmap to pluralistic alignment, specifically using large language models as a test bed. We identify and formalize three possible ways to define and operationalize pluralism in AI systems: 1) *Overton pluralistic* models that present a spectrum of reasonable responses; 2) *Steerably pluralistic* models that can steer to reflect certain perspectives; and 3) *Distributionally pluralistic* models that are well-calibrated to a given population in distribution. We also formalize and discuss three possible classes of *pluralistic benchmarks*: 1) *Multi-objective* benchmarks, 2) *Trade-off steerable* benchmarks that incentivize models to steer to arbitrary trade-offs, and 3) *Jury-pluralistic* benchmarks that explicitly model diverse human ratings. We use this framework to argue that current alignment techniques may be fundamentally limited for pluralistic AI; indeed, we highlight empirical evidence, both from our own experiments and from other work, that standard alignment procedures might *reduce* distributional pluralism in models, motivating the need for further research on pluralistic alignment.

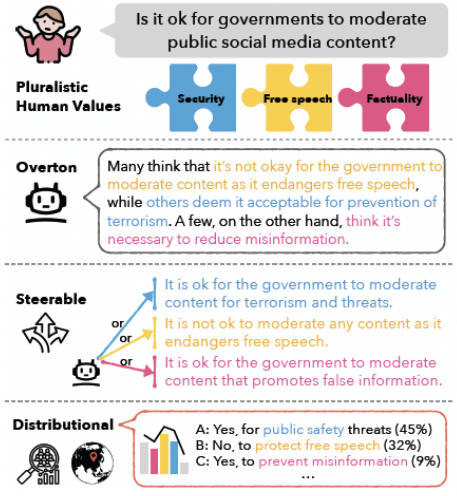


Figure 1. Three kinds of pluralism in models.

1. Introduction

AI alignment aims to ensure that a system works with human intentions and values (Leike et al., 2018; Ji et al., 2024; Gabriel, 2020). However, even within a single task or prompt, individual people vary widely in their goals, intentions, and values. As a broader set of people use and rely upon AI systems, we need systems that can understand and cater to a broader set of needs. In other words, we need systems that are *pluralistic*, or capable of representing a diverse set of human values and perspectives. While many in the community have argued for this (Bai et al., 2022b; Gordon et al., 2022; Sorensen et al., 2023), at least two important questions remain: *How, concretely, can a system be pluralistic?* and *How might benchmarks be designed to measure pluralism*?

In this piece, we advocate for explicit pluralistic considerations in aligning AI systems (§2). In particular, we

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Whose Truth? Pluralistic Geo-Alignment for (Agentic) AI

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ABSTRACT

AI alignment describes the challenge of ensuring (future) AI systems behave in accordance with societal norms, values, and goals. Alignment is now central to research on foundation models and AI agents. Most recent work focuses on methods to prevent potentially harmful biases, account for social inequalities, improve AI safety, and enhance explainability. Notably, the debiasing 'corrections' applied to various stages of AI/ML workflows may lead to outcomes that diverge strongly from current statistical realities on the ground. For instance, text-to-image models may depict a balanced gender ratio of company leadership, despite existing imbalances. However, an often overlooked dimension is the *geographic variability* of alignment. What is considered appropriate, truthful, or legal can vary greatly between regions due to cultural differences, political realities, or legislation. Hence, some model outputs align without further knowledge of the user's geospatial context, while others are highly sensitive to it. Put differently, whether these outputs align varies geographically. E.g., statements about Kashmir cannot be generated without understanding the user's origin and current location. From a common-sense perspective, this problem is hardly new. In fact, Google Maps will render different administrative borders based on the user's location. Interestingly, in both knowledge representation and representation learning, spatiotemporal context, e.g., due to the monotonic nature of reasoning, remains a major challenge. Until very recently, these were largely theoretical problems. What is truly novel is the scale and level of automation at which AI systems now mediate knowledge, express opinions, and represent *reality* to millions of users across borders, often with little transparency or oversight regarding how context is handled. With agentic AI on the horizon, the urgency for pluralistic, geographically aware alignment, rather than one-size-fits-all solutions, is growing. Here, we motivate and formalize the **vision of geo-alignment**, outline how it goes beyond pluralistic alignment by offering learnable spatially explicit patterns, and suggest concrete avenues for future research.

CCS CONCEPTS

• **Social and professional topics** → **Geographic characteristics**; • **Computing methodologies** → **Philosophical/theoretical foundations of artificial intelligence**; • **Applied computing**;

KEYWORDS

GeoAI, Geo-Alignment, Pluralistic Alignment, AI Ethics, Bias

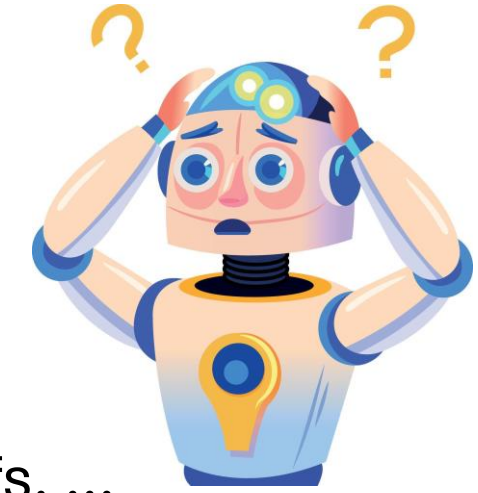
1 INTRODUCING GEO-ALIGNMENT

In a nutshell, *alignment* asks the question of *how to ensure AI systems behave according to societal norms, values, and goals, even though those goals may themselves be complex, contested, context-dependent, or even conflicting*. A system is *aligned* if its actions advance these norms, values, and goals, and *misaligned* if it acts otherwise [26, 35]. The challenge of *super-alignment* goes one step further by focusing on (potential) future super-intelligence. This not only changes the urgency, but also the scope and methods, as such systems may reason beyond our abilities, pursue goals we cannot comprehend, and act at speeds that render post-hoc oversight impossible [6].

Unfortunately, however, established research and practice often glosses over the fact that the goals of individuals, groups, and societies do not always converge. Put differently, **existing work risks confusing inter-human variation with noise** [29]. This was acceptable as long as AI alignment was largely a philosophical endeavor, but how is an AI agent supposed to align if we humans cannot? Hence, AI research has begun to account for *value pluralism* [5] and *pluralistic alignment* [29]. For example, the authors distinguish between (1) *overton pluralism*, where all potentially 'reasonable' answers are returned; (2) *steerable pluralism*, where the model outputs are guided towards a given perspective (called an attribute); and (3) *distributional pluralism*, where outputs are drawn in accordance with those of a desired reference population.

Geography, as defined by the interplay of its spatial and temporal structure through dependence and heterogeneity, and the societal layer arising on top, has not been prominently featured in alignment research so far. This is a problem because it risks AI systems producing outputs that are potentially locally inappropriate, culturally insensitive, or even illegal. For example, a system may

Understanding Pluralism and Alignment



- Pluralism
  - Coexistence of diverse groups in society
  - Diverse values, perspectives, cultures, religions, political beliefs, …
- Alignment
  - AI-Systems should behave according to societal norms, values and goals



A Roadmap to Pluralistic Alignment Problems

- Multiple reasonable answers → model response is only one of them

- Models may be biased towards certain perspectives
  - White, male, liberal, wealthy, american

- Evaluation benchmarks do not properly represent these issues

A Roadmap to Pluralistic Alignment Research

- Comparison of pre- and post-aligned models
- Three definitions of pluralistic models
- Three definitions of pluralistic benchmarks

A Roadmap to Pluralistic Alignment
Key Findings

- Alignment can reduce response diversity
- Pluralism needs different training and evaluation
- Catering to the average opinion does not represent the population

Whose Truth? Pluralistic Geo-Alignment for AI Problem

- If an answer is reasonable depends on the location, model answers often ignore this

⟶ Geo-Alignment

  - Adding learnable spatially explicit patterns

Whose Truth? Pluralistic Geo-Alignment for AI Research

- Formalizing Geo-Alignment

- Highlighting some proposed research directions

Whose Truth? Pluralistic Geo-Alignment for AI Key Findings

- Geo-Alignment can be formalized using a divergence measure

- Spatial variation is not properly implemeted in AI alignment but offers learnable solutions and should be researched further

Experiments for our Presentation

- Create questions with multiple correct answers that differ between places
- Collect answers from commen AI models, answered from different perspectives
- Collect answers from the audience
- Compare, maybe build a small application to make it pretty



THANK YOU FOR YOU ATTENTION!